



<Customer Full Name> Oracle Base Database Cloud@Customer

Solution Definition

08-09-2026| Version 1.0
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- Guide:
- Author Responsibility
- Chapter 1-3: Sales Consultant
 - Chapter 4: Implementer

Document Control

Guide:

The first chapter of the document describes the metadata for the document. Such as versioning and team members.

1.1 Version Control

Guide:

A section describing the versions of this document and its changes.

Example:

Version	Authors	Date	Comments
1.0	Author name	10 th Aug 2026	First release

1.2 Team

Guide:

A section describing the team.

Example:

Name	Email	Role	Company
Author #1	example@example.com	DBP Account Cloud Engineer, Author	Oracle
Author #2	example@example.com	Database-C@C BB, Contributor	Oracle

1.3 Abbreviations List

ACME	A Company that Makes Everything (Global Bank in this document)
OCI	Oracle Cloud Infrastructure
VCN	Virtual Cloud Network
VM	Virtual Machine
AD	Availability Domain
FD	Fault Domain
DR	Disaster Recovery
DBaaS	Database as a service
MAA	Maximum Availability Architecture
RAC	Real Application Clusters
RMAN	Oracle Recovery Manager
OKV	Oracle Key Vault
DNS	Domain Name Server

Business Context

PKI	Public Key Infrastructure
BOM	Bill of Material
AES	Advanced Encryption Standard
DMZ	de-militarized zone
BaseDB-C@C	Base Database Service on Cloud@Customer
BaseDB-C@C	Base Database Database Service on Cloud@Customer
ZDLRA	Zero Data Loss Recovery Appliance
RPO	Recovery Point Objective
RTO	Recovery Time Objective

1.4 Document Purpose

Guide:

Describe the purpose of this document and the Oracle-specific terminology, specifically around 'Workload'.

Example:

This document provides a high-level solution definition for the Oracle solution and aims at describing the current state, and to-be state as well as a potential high-level project scope and timeline for <Service Provider>.

The document may refer to a 'Workload', which summarizes the full technical solution for a customer (You) during a single engagement. The Workload is described in the chapter Workload Requirements and Architecture.

This is a living document. Additional sections will be added as the engagement progresses resulting in a final Document to be handed over to the <Service Provider>.

Business Context

Guide:

Describe the customer's business and background. What is the context of the customer's industry and LoB? What are the business needs and goals that this Workload is an enabler for? How does this technical solution impact and support the customer's business goals? Does this solution support a specific customer strategy, or maybe certain customer values? How does this solution help our customers to either generate more revenue or save costs?

Consider the different context of a new deployment of Base Database Cloud@Customer vs. an expansion of an existing deployment.

2.1 Executive Summary

Guide:

A section describing the Oracle differentiator and key values of the solution of our solution for the customer, allowing the customer to make decisions quickly.

2.2 Workload Business Value

Guide:

A clear statement of specific business value as part of the full workload scope. Try to keep it SMART: Specific, Measurable, Assignable, Realistic, and Time-Related - Agree on the business value with the customer. Keep it business-focused and speak the language of the LoB which benefits from this Workload: "Increase Customer Retention by 3% in the next year" or "Grow Customer Base with Executive Decision-Making from our Sales and Support Data". Avoid technical success criteria such as "Migrate App X to Oracle Cloud" or "Provision 5 Compute Instances". Avoid Oracle success criteria and language "Get Workload Consuming on OCI".

Workload Requirements and Architecture

3.1 Overview

Guide:
Describe the Workload: What applications and environments are part of this Workload migration or new implementation project, and what are their names? The implementation will be scoped later and can be a subset of the Solution Definition and proposed overall solution. For example, a Workload could exist of two applications, but the implementer would only include one environment of one application. The workload chapter is about the whole Workload and the implementation scope will be described later in the chapter Solution Scope.

3.2 <Customer> Non-Functional Requirements

Guide:
Describe the high-level technical requirements for the Workload. Consider all sub-chapters, but decide and choose which Non-Functional Requirements are necessary for your engagement. You might not need to capture all requirements for all sub-chapters.
This chapter is for describing customer-specific requirements (needs), not to explain Oracle solutions or capabilities.

3.2.1 Regulations and Compliances Requirements

Guide:
This section captures specific regulatory or compliance requirements for the Workload. These may limit the types of technologies that can be used and may drive some architectural decisions.
The Oracle Cloud Infrastructure Compliance Documents service lets you view and download compliance documents: <https://docs.oracle.com/en-us/iaas/Content/ComplianceDocuments/Concepts/compliancedocoverview.htm>.
If there are none, then please state it. Leave the second sentence as a default in the document.

Example:
At the time of this document creation, no Regulatory and Compliance requirements have been specified.
In addition to these requirements, the CIS Oracle Cloud Infrastructure Foundation Benchmark, v1.2 will be applied to the Customer tenancy.

3.2.2 Environments

Guide:
A diagram or list detailing all the required environments (e.g. development, test, live, production, etc.).
If you like to describe a current state, you can use or add the chapter 'Current State Architecture' before the 'Future State Architecture'.

Example:
There are three environments that needs to be covered in the solution.

Name	Location	Data Center	DR	High Availability	RPO	RTO
Production	Budapest	DC1 and DC2	Yes	Yes	~0	2 hrs
Non-Production	Budapest	DC1 or DC2	No	Yes	24 hrs	24 hrs
Miscellaneous	Budapest	DC1 only	No	Yes	4 hrs	12 hrs

3.2.3 High Availability and Disaster Recovery Requirements

Guide:
This section captures the resilience and recovery requirements for the Workload. Note that these may be different from the current system.
The Recovery Point Objective (RPO) and Recovery Time Objective (RTO) requirement of each environment should be captured in the environments section above, and wherever possible.

- What is the RTO and RPO requirements of the Application?

Commented [MV1]: text=test
Commented [MV2]: Sate=State

- What are the SLAs of the application?
- What are the backup requirements?

Note that if needed, this section may also include an overview of the proposed backup and disaster recovery proposed architectures.

Example:
Oracle MAA Best Practices and Reference Architectures should be considered for all environments.
All production databases need to be highly available and covered by a disaster recovery solution across the two data centers in Stockholm, Your Data Center 1 and Your Data Center 2. Performance needs to be maintained during failure of components or systems.
Databases in all environments should be unaffected by maintenance.

3.2.4 Backup and Recovery requirements

Guide:
This section captures the backup and recovery requirements for the Workload. Note that these may be different from the current system.
The Recovery Point Objective (RPO) and Recovery Time Objective (RTO) requirement of each environment should be captured in the environments section above, and wherever possible.

- What is the RTO and RPO requirements for the backup and recovery?
- What is the backup retention requirements

Example:
Current backup solution with NFS shares should be used initially, but the solution needs to be able to be upgraded with Zero Data Loss Recovery Appliance to accommodate increasing demands for Ransomware Protection (Compliance and Immutability) and Zero RPO.

Environment	RPO	RTO
Production	~0	2 hrs
Non-Production	24 hrs	24 hrs
Miscellaneous	4 hrs	12 hrs

3.2.5 Data Center requirements

Guide:
Capture which data centers is going to be used. The Base Database Cloud@Customer is designed for rack installation and includes two servers and one storage shelf.
The electrical requirements can be found in the documentation:
<https://docs.oracle.com/en/cloud/cloud-at-customer/base-database/admin/prep-site.html#electrical-power-requirements>

The network connection requirements can be collected in the table below. Please note that the Client and Backup network connections must be the same type (copper or optical). For CPS connectivity only copper connections are supported.

Data Center <NICKNAME >	Number of connections	Speed (10 or 10/25 Gbps)	Cabling (optical or copper)
Client network	4		
Backup network (optional)	4 (optional)		
Control Plane Server (CPS) network	4	10 Gbps	copper

Example:

Workload Requirements and Architecture

Current data centers (Your Data Center 1 and Your Data Center 2) must be used because of data residency requirements and application (to database) latency requirements. The databases / database services need to adhere to the data center requirements present.

Your Data Center 1	Number of connections	Speed (10 or 10/25 Gbps)	Cabling (optical or copper)
Client network	4	10 Gbps	Optical
Backup network (optional)	4 (optional)	10 Gbps	optical
Control Plane Server (CPS) network	4	10 Gbps	Copper

Your Data Center 2	Number of connections	Speed (10 or 10/25 Gbps)	Cabling (optical or copper)
Client network	4	10 Gbps	Optical
Backup network (optional)	4 (optional)	10 Gbps	optical
Control Plane Server (CPS) network	4	10 Gbps	Copper

3.2.6 Security Requirements

Guide:

Capture the Non-Functional Requirements for security-related topics. The requirements can be (but don't have to be) separated into:

- Identity and Access Management
- Data Security

Other security topics, such as network security, application security, or others can be added if needed.

Make sure to capture the requirements on Key Management as TDE (Transparent Data Encryption) is mandatory on all databases in Base Database Cloud@Customer.

BaseDB-C@C needs to connect to OCI Region so capture security requirements for this connection. For example, if Fast Connect or Site-to-Site VPN is required. For a new deployment this would need to be configured prior to Data Center Site Inspection and IP Checks.

Example:

Site-to-Site encryption required, security policies do not allow for public internet connections. This can either be Site-to-Site VPN or preferably a dedicated and private connection.

The existing installation of Oracle Key Vault (OKV) must be used.

3.2.7 Networking Requirements

Guide

Capture the Non-Functional Requirements for networking-related topics. You can use the networking questions in the Annex As businesses increasingly rely on Cloud Infrastructure to store, process, and transmit sensitive data, the need for comprehensive security solutions has never been more important. Potential customers evaluating network security solutions typically prioritize the following requirements: Some of the broader category considerations are below.

- Data Protection: Safeguarding sensitive information against unauthorized access, theft, or modification is a primary concern for any organization and industry today.

Workload Requirements and Architecture

- *Threat Prevention: Advanced capabilities like IDPS and malware detection for blocking threats.*
- *Data Loss Prevention (DLP): Monitoring and controlling sensitive data transmission.*
- *Encryption and Decryption: Inspecting encrypted traffic without compromising privacy.*
- *Threat Prevention: Proactively identifying and mitigating security threats is essential for maintaining the integrity of network infrastructure.*
 - *Intrusion Detection and Prevention: Monitoring for suspicious or malicious activity.*
 - *Application Control: Granular control over specific applications or services.*
 - *URL Filtering: Controlling access to permitted URLs.*
- *Security compliance: Does your organization have network security requirements based on industry or organization compliance? For example - SAMA (Saudi Arabia Monetary Authority), HIPAA (Health Insurance Portability and Accountability Act), GDPR (General Data Protection Regulation), SWIFT, etc.*

Try to capture network infrastructure requirements in the Data Center requirements section above.

Example:

Site-to-Site encryption is required, security policies do not allow for public internet connections. This can either be a private Site-to-Site VPN or preferably a dedicated connection like Fast Connect.

3.2.8 Management and Monitoring

Guide:

This subsection helps you capture any requirements for customer management and monitoring needs - e.g. system monitoring, systems management, log analysis, etc.

When you move or start an OCI project, you have a choice to use the tools you are familiar with (should they support modern application architectures), replace them with OCI native Observability services, or use a combination to improve your visibility.

When contemplating how to proceed, here are some general questions that will guide you:

- *Does the tool manage across hybrid and multi-cloud environments?*
- *What is the cost of integrating the existing tool with OCI?*
- *Is my current monitor tool enabling you to prevent issues versus reacting to them?*
- *Does the tool tell you how much impact there has been on users or just that there was an impact like something is down or unavailable?*
- *Does the tool provide the full vision of applications and their infrastructure or just a piece of them or specific technology?*

Example:

Task	Target	Location	New	Notes
Application Monitoring	All targets	On-Prem and OCI	No	
Monitoring	All targets	OCI (Migration)	No	
Log Management	All targets	OCI (Migration)	No	
Insight	All Oracle DBs	OCI (Migration)	No	

3.3 Future State Architecture

Guide:

The Workload Future State Architecture can be described in various forms. In the easiest case, we describe a Logical Architecture, possibly with a System Context Diagram. A high-level physical architecture is mandatory as a description of your solution.

This should be the final architecture as part of the pre-sales solution, not an intermediate or draft version

3.3.1 Mandatory Security Best Practices

Workload Requirements and Architecture

Guide:

Use this text for every engagement. Do not change. Aligned with the Cloud Adoption Framework

The safety of <Customer Name>'s Oracle Cloud Infrastructure (OCI) environment and data is the <Customer Name>'s priority. The following table of OCI Security Best Practices lists the recommended topics to provide a secure foundation for every OCI implementation. It applies to new and existing tenancies and should be implemented before the Workload defined in this document will be implemented.

Workload-related security requirements and settings like tenancy structure, groups, and permissions are defined in the respective chapters.

Any deviations from these recommendations needed for the scope of this document will be documented in the chapters below. They must be approved by <Customer Name>.

<Customer Name> is responsible for implementing, managing, and maintaining all listed topics.

CATEGORY	TOPIC	DETAILS
User Management	IAM Default Domain	Multi-factor Authentication (MFA) should be enabled and enforced for every non-federated OCI user account. - For configuration details see Managing Multi-Factor Authentication. In addition to enforcing MFA for local users, Adaptive Security will be enabled to track the Risk Score of each user of the Default Domain. - For configuration details see Managing Adaptive Security and Risk Providers.
	OCI Emergency Users	A maximum of three non-federated OCI user accounts should be present with the following requirements: - Username does not match any username in the Customer's Enterprise Identity Management System - Are real humans. - Have a recovery email address that differs from the primary email address. - User capabilities have Local Password enabled only. - Has MFA enabled and enforced (see IAM Default Domain).
	OCI Administrators	Daily business OCI Administrators are managed by the Customer's Enterprise Identity Management System. This system is federated with the IAM Default Domain following these configuration steps: - Federation Setup - User Provisioning - For configuration guidance for major Identity Providers see the OCI IAM Identity Domain tutorials.
	Application Users	Application users like OS users, Database users, or PaaS users are not managed in the IAM Default Domain but either directly or in dedicated identity domains. These identity domains and users are covered in the Workload design. For additional information see Design Guidance for IAM Security Structure.
Cloud Posture Management	OCI Cloud Guard	OCI Cloud Guard will be enabled at the root compartment of the tenancy home region. This way it covers all future extensions, like new regions or new compartments, of your tenancy automatically. It will use the Oracle Managed Detector and Responder recipes at the beginning and can be customized by the Customer to fulfill the Customer's security requirements. For configuration details see Getting Started with Cloud Guard. Customization of the Cloud Guard Detector and Responder recipes to fit the Customer's requirements is highly recommended. This step requires

CATEGORY	TOPIC	DETAILS
		thorough planning and decisions to make.
	OCI Vulnerability Scanning Service	- For configuration details see Customizing Cloud Guard Configuration In addition to OCI Cloud Guard, the OCI Vulnerability Scanning Service will be enabled at the root compartment in the home region. This service provides vulnerability scanning of all Compute instances once they are created. - For configuration details see Vulnerability Scanning.
Monitoring	SIEM Integration	Continuous monitoring of OCI resources is key for maintaining the required security level (see Regulations and Compliance for specific requirements). See Design Guidance for SIEM Integration to implement integration with the existing SIEM system.
Additional Services	Budget Control	OCI Budget Control provides an easy-to-use and quick notification on changes in the tenancy's budget consumption. It will be configured to quickly identify unexpected usage of the tenancy.
		For configuration details see Managing Budgets

3.3.2 Naming Conventions

Guide:

This chapter describes naming convention best practices and usually does not require any changes. If changes are required please refer to [Landing Zone GitHub](#). The naming convention zone needs to be described in the Solution Design by the service provider.

Use this chapter ONLY for new cloud deployments and remove it for brownfield deployments.

Naming convention is an important part of any deployment to ensure consistency, governance, and security within your tenancy. Find [here](#) Oracle's recommended best practices.

3.3.3 OCI Landing Zone Solution Definition

Guide:

This chapter describes landing zone best practices and usually does not require any changes. If changes are required, please refer to [Landing Zone GitHub](#). The full landing zone needs to be described in the Solution Design by the service provider.

Use this template ONLY for new cloud deployments and remove it for brownfield deployments.

Example:

An OCI Landing Zone sets the foundations for a secure tenancy, providing design best practices and operational control over OCI resources. A Landing Zone also simplifies the onboarding of workloads and teams, with clear patterns for network isolation and segregation of duties in the organization, which sets the cloud operating model for day-to-day operations. Oracle highly recommends the use of an OCI Landing Zone for any deployment. Use these guidelines to set up your OCI Landing Zone, including design considerations, approaches, and solutions to use.

Note that all workloads in a tenancy should sit on top of a Landing Zone, meaning that the workload architecture defined in the next section can be subject to adjustments (e.g., network structure) towards the landing zone model, along with other future workloads.

3.3.4 Logical Solution Architecture

Guide:

Provide a high-level logical Oracle solution for the complete Workload. Indicate Oracle products as abstract groups, and not as physical detailed instances. Create an architecture diagram following the latest notation and describe the solution.

To implement a solution the Physical Architecture is needed in the next chapter. The physical notation can show individual components with physical attributes such as IP addresses, hostnames, or sizes.

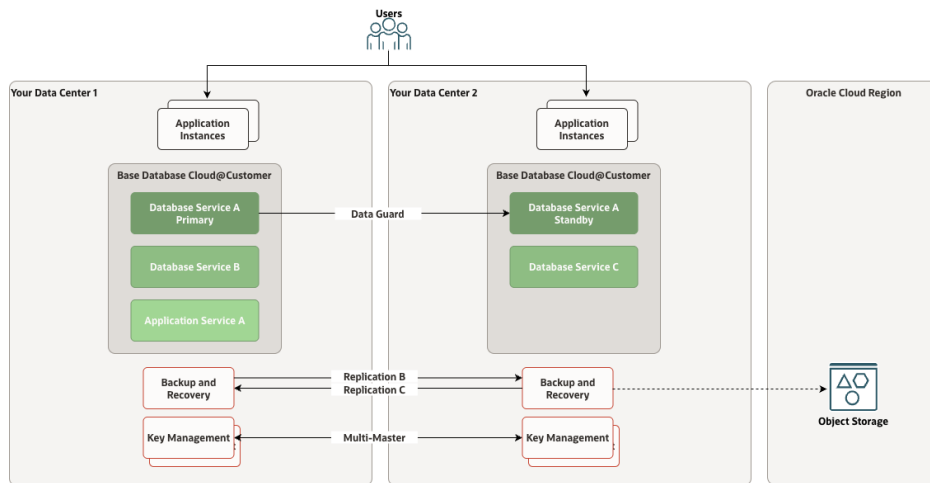
The Oracle Cloud Notation, OCI Architecture Diagram Toolkits

Example:

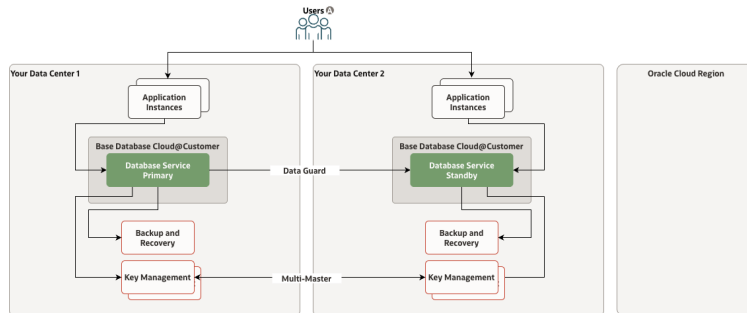
Database consolidation with multiple service levels.

Based on requirements for data residency and local application performance this solution delivers Database as a Service (DBaaS) in Your Data Centers using Base Database Cloud@Customer. Considering Oracle Maximum Availability Architecture (MAA) multiple service levels have been defined in the logical solution architecture below.

The logical architecture have multiple service levels for Production (Database Service A), non-Production divided between data centers (Database Service B and C) and Miscellaneous (Database Service D) due to ranging requirements.



3.3.4.1 Service Level with High Availability, Disaster Recovery and Data Protection



Details of Database Service A based on MAA Reference Architecture "Gold".

3.3.5 Physical Architecture

Guide:

The Workload Architecture is typically described in a physical form. This should include all solution components. You do not have to provide solution build or deployment details such as IP addresses.

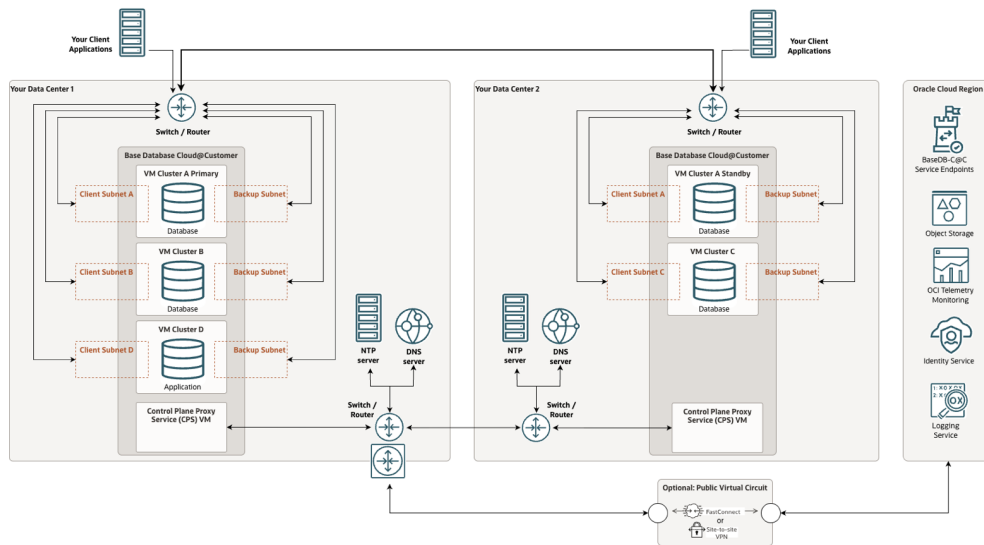
Please describe the solution with an architecture image plus a written text. If you have certain specifics you like to explain, you can also use the Solution Consideration chapter to describe the details there.

Example:

Database consolidation with multiple service levels.

Based on the defined logical architecture the physical architecture to implement consists of two Base Database Cloud@Customer systems, one in each of Your Data Centers. To implement the required MAA Reference Architecture service levels there will be three VM Clusters in BaseDB-C@C in Your Data Center 1 and two VM Clusters will be created in BaseDB-C@C in Your Data Center 2.

Workload Requirements and Architecture



The VM Clusters fulfill the following requirements:

- Network segmentation due to security.
Each VM Cluster have its own Client Subnet including VLAN ID.
- Workload isolation
Each VM Cluster have its own dedicated resources in CPU, Memory and Storage Capacity.
- Application / User separation
Each VM Cluster can have its own users and groups as defined in the Identity Service.
- Required Service Level Objectives (for example MAA Gold, Silver)
Each VM Cluster is designed against a given MAA Reference Architecture.

3.3.6 Key Solution Proposal Components

Our proposal is based on the following main technical components:

3.3.6.1 Base Database Cloud@Customer (BaseDB-C@C)

Guide:

This text is generic and should be kept as is, please change the <CUSTOMER NAME>

Example:

Oracle Base Database Cloud@Customer brings the performance, automation, and economics of Oracle Database Service

Workload Requirements and Architecture

managed by Oracle to data centers in <CUSTOMER NAME>. This is the easiest way for <CUSTOMER NAME> to start using cloud database resources in their data centers while meeting strict data residency standards. Base Database Cloud@Customer incorporates unique optimizations that enable databases to run faster with less management and at lower costs, to better leverage <CUSTOMER NAME> data.

Value Proposition

Oracle's value proposition to support <CUSTOMER NAME>'s needs is built around the Base Database Cloud@Customer cloud platform:

Resilience and Availability

- Modern, highly available, and secure data architecture by design
- Services installed in <CUSTOMER NAME> data center, dedicated and isolated, to meet data sovereignty requirements
- Simplified patch maintenance and proactive threat management

Performance, agility, and flexibility

- Low latency to keep data as close as possible to applications and choose the timeline for modernizing these databases as needed
- Ability to scale the platform as needs change without increasing complexity
- Maximum availability architecture (e.g. PROD) with fast RTO and RPO, service continuity, etc.)
- Obsolescence and risks
- Simplify and standardize to enhance security
- Automation to achieve more predictable service levels and address a wide range of security vulnerabilities Security
- Encryption of all data at rest and in transit, with BP2i control of encryption keys.
- Ability to use an existing HSM, ensuring that <CUSTOMER NAME> has data privacy, and Oracle Key Vault between that HSM and the databases, to have the full level of integration offered by the provisioning and management tools of the Base Database Cloud@Customer architecture

Commented [MV3]: PRA??

Note: OKV is fully integrated with the Base Database Cloud@Customer solution and enables the use of all management APIs. Because these are maintained by Oracle, this causes a decrease in TCO. By using OKV, Oracle provides support and onboarding only, without the need for a third-party contact. From a long-term perspective, OKV is the only possible solution to date for outsourced key management for Autonomous Database

Speed of implementation / Time to Market

- Time to Market optimization is made possible thanks to the comprehensive "Infrastructure as Code" approach, which allows all environments to be managed through a DevOps methodology to ensure repeatability.
- Implementation of applications directly and without the need for infrastructure teams, greatly accelerating their implementation and simplifying their maintenance.

Access to innovative services and a unified data platform

Innovation around data services has been at the heart of Oracle's DNA since its inception. Beyond access to all cloud services around data, <CUSTOMER NAME> will be able to benefit from innovation around the Oracle database. Beyond modernization, some use cases are potentially better served by the direct use of the PaaS service available on the Base Database Cloud@Customer platform, with a project time potentially shorter or equivalent to reproducing the existing or not benefiting from the PaaS services

3.3.6.2 Backup and Recovery Options

Guide:

This text is generic and should be kept as is, please change the <CUSTOMER NAME>

Workload Requirements and Architecture

Example:

The Logical Architecture describes the high-level backup and recovery solution.

Base Database Cloud@Customer offers two approaches to configure and take backups as a recommended solution: Oracle Managed Backup and User Configured Backup.

- Oracle Managed Backup: Database backups are managed entirely by the Base Database Cloud@Customer service based on a one-time configuration.
- User Configured Backup: It is the user's responsibility to configure and execute backup operations according to their preferences.

Customers can also choose from different Backup Destinations

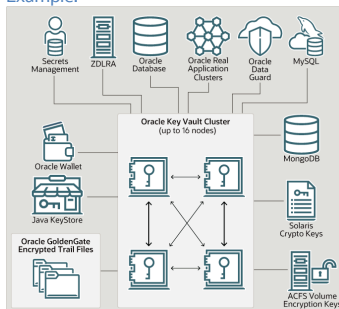
- Customer Managed Zero Data Loss Recovery Appliance backup destination
- Customer Managed NFS backup destination

3.3.6.3 Oracle Key Vault

Guide:

This text is generic and should be kept as is, please change the <CUSTOMER NAME>

Example:



Oracle Key Vault is engineered to deliver performant, fault-tolerant, and flexible encryption key management for transparent data encryption (TDE) as part of Oracle Advanced Security. Key Vault works with all database deployment options, including Oracle Real Application Clusters (Oracle RAC), Oracle Data Guard, sharded databases, RMAN backups, and Oracle Multitenant pluggable databases. Key Vault scales to support highly consolidated workloads on engineered systems, such as Oracle Base Database, Base Database Database Service on Dedicated Infrastructure (BaseDB-D) as well as full integration into the database provisioning workflow for Base Database Cloud@Customer (BaseDB-C@C).

We can therefore provide <CUSTOMER NAME> with assurance that Oracle Databases will be deployed on the best environment to ensure the availability and sustainability of critical data and to meet European requirements for regulating and resuming banking activities in the event of cyber-attacks.

3.4 Solution Considerations

Guide:

Describe certain aspects of your solution in detail. What are the security, resilience, networking, and operations decisions you have taken that are important for your customer?

3.4.1 High Availability and Disaster Recovery

Guide:

- Resilience on OCI.

Workload Requirements and Architecture

- Workload Related Content.
- What happens when site 1 is down?
- What happens when XYZ?
- How is HA achieved?

Example:

Customer requires to adhere to Oracle MAA principles.

Following has been decided:

- Production Databases:
 - Database Service A
 - MAA Gold with 2-node dedicated VM-cluster in DC1 and 2-node dedicated VM-Cluster in DC2
 - Full RAC
 - Data Guard with Your DC1 as primary DC and Your DC2 as standby DC
 - Backups will be made from the local instance to the local backup environment.
- Non-Production Databases:
 - Database Service B & C
 - MAA Silver with 2-node VM-Cluster in Your DC1 for Database Service B and 2-node VM-Cluster in Your DC2 for Database Service C
 - Database Service B and Database Service C spread across DC for efficiency.
 - Full RAC
 - Local backups that will be replicated through the backup solution to your other DC.

3.4.2 Backup and Recovery

Guide:

Base Database Cloud@Customer requires a backup network, with the intent to run Automatic or Manual backups of databases. Describe the intended backup solution.

Example:

Customer has standardized backups for Oracle DB based on custom RMAN scripts sending DB backups to external NFS share. In the first phase DB backups from the BaseDB-C@C need to be integrated with and stored on the external NFS share(s). Customer decided to work with Oracle Managed Backups on the BaseDB-C@C instead of his own custom scripts. Customer requirement is to have 60-day retention for Database Service A (Production) and 30-day retention for Database Service B&C (Non-Production). Weekly full backup and daily incremental backups will be taken, full backups on Sunday, incremental backups the other days of the week. Every 30 min backups of the archive logs will be taken as well. Backup retention will be configured per customer's requirement. Backups will be stored on the external NFS share(s) that the customer will continue to manage. In a later phase, to meet the RPO close to zero requirement, Oracle will work with the customer to evaluate and build a case around Oracle Zero Data Loss Recovery Appliance.

3.4.3 Security

Please see our security guidelines in the Annex.

3.4.4 Networking

Guide:

A list of possible Oracle solutions can be found in the Annex.

Example:

In Datacenter Networking

Workload Requirements and Architecture

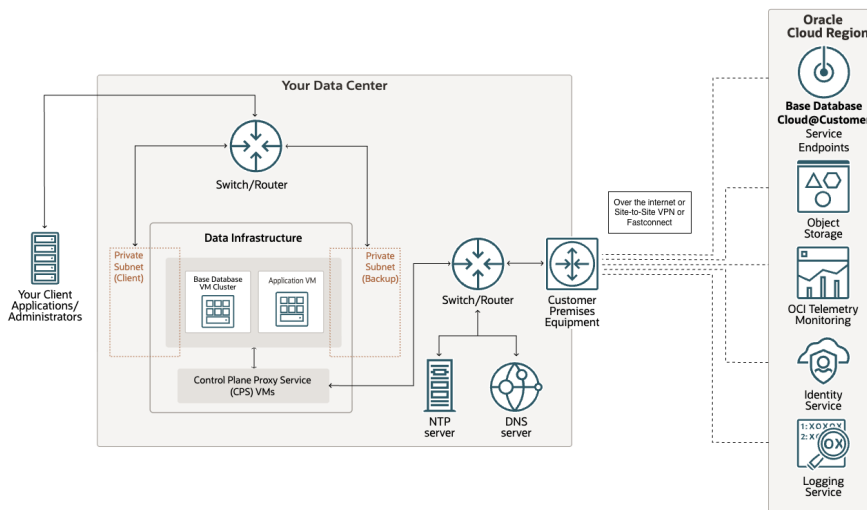
All networking within the customer's datacenter is optical multimode and the target Base Database Cloud@Customer will connect to the core network. Appropriate transceivers (SFP+/SFP28 modules) for the network ports on the BaseDB-C@C will be part of delivery if fiber connectivity option was selected for the client and backup network.

Transceivers for customer ports, patch panels and optical network fibers will be delivered by customer.

Network connectivity to OCI

Customer does not allow a connection over public internet to OCI. Therefore, it has been decided to deploy a FastConnect over a private circuit to provide the required CPS connectivity.

Networking requirement for Base Database Cloud@Customer documentation link.



3.4.5 Manageability and Observability

Guide:

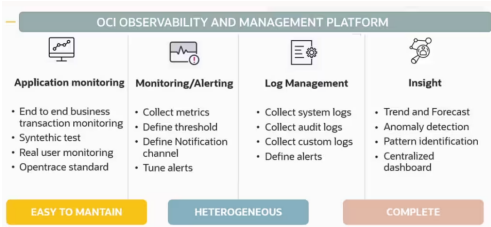
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Observability is a technology advancement focused on getting insights from a vast array of data, logs, and events generated within an IT environment. By implementing an Observability strategy, organizations gain the capability to anticipate system disruptions, prevent resource overconsumption, and enhance the overall application user satisfaction. That means being proactive, which is a must, especially in a distributed environment.

Gone are the days when the IT landscape remained a mysterious black box. The company's digitalization and the Cloud model compel C-level executives to gain comprehensive insights into asset utilization. The efficient allocation of resources directly influences budgetary considerations.

Observability helps organizations examine how well their infrastructure is working, predict future needs, and help take proactive steps to improve efficiency and protect investments. Therefore, Observability tools are needed to cover these important areas.

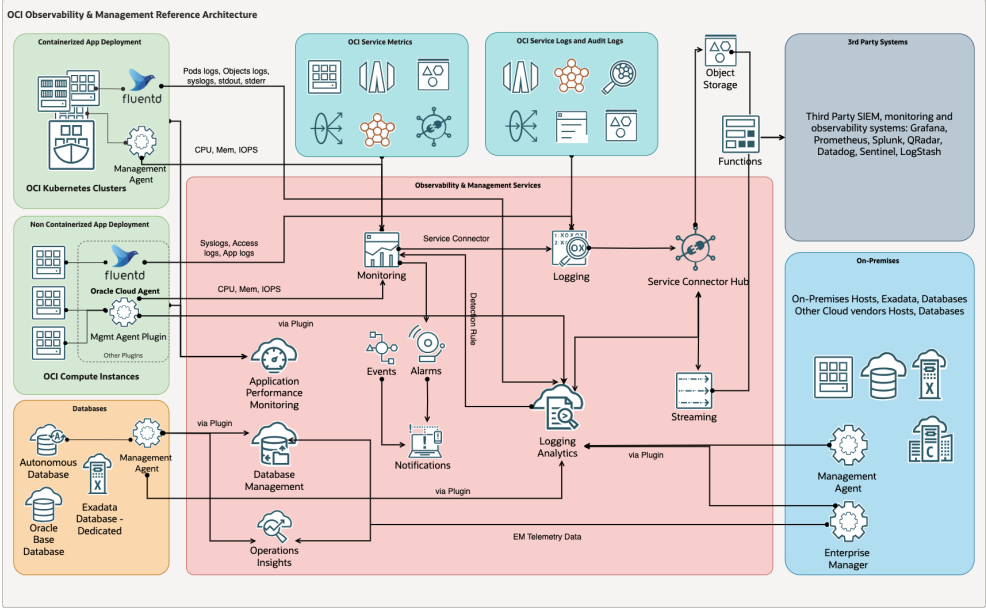
Workload Requirements and Architecture



Observability and Manageability

3.4.5.1 Observability Architecture

The basic monitoring OCI services collect the data and send logs and metrics to OCI Monitoring and Logging services. If you want to apply machine-learning capabilities and perform analysis, you can send the data to the Logging Analytics service. If you want to use OCI Logging Analytics to collect logs coming from both on-premises and cloud sources to analyze them for auditing, security purposes, or to integrate data with an external SIEM solution, the Connector Hub serves as the solution. It's advisable to plan your monitoring strategy by considering both the O&M (Observability and Management) native service of OCI and its integration with third-party tools, as O&M is flexible and a highly customizable solution.



OCI Architecture

3.4.5.2 Real-Time Monitoring

Real-time monitoring is the delivery of continuously updated data about systems, processes, or events. Such monitoring provides information streaming at zero or low latency, so there is minimal delay between data collection and analysis. It enables quick detection of anomalies, performance issues, and critical events.

Please find all references for this chapter in the Annex.

3.4.5.3 Performance and Tuning

Performance tuning is the improvement of system performance. It can be done proactively to prevent issues or reactively in response to increased workload, which is crucial for avoiding system outages.

Please find all references for this chapter in the Annex.

3.4.5.4 Administration

Administrator tasks involve upholding a data management policy and ensuring essential equipment functionality, such as instance management, backup & restore operations, key management, and allocating resources from the database to the storage.

Please find all references for this chapter in the Annex.

3.4.5.5 Troubleshooting

Issues can happen on several levels. To identify the root cause, it is important to be able to correlate resources, drill down into the issues, and analyze trends in the systems. It's crucial to consider that the application itself might be the root cause of the issue. Therefore, it's essential to gather information about the application's behavior and performance to fully understand the problem and resolve it effectively. Troubleshooting also allows you to avoid an outage which is why it is important to notice issues as early as possible.

Please find all references for this chapter in the Annex.

3.4.5.6 Cost Control and Chargeback

Cost control is the practice of identifying and reducing business expenses to increase profits. It starts with the budgeting process. Cost control is an important factor in maintaining and growing profitability.

IT chargeback can provide greater visibility into the costs of IT services and infrastructure usage. It enables organizations to identify opportunities for cost optimization and reduce wasteful spending.

Cost control and chargeback are critical concerns, especially for companies transitioning to the cloud, presenting new financial operational challenges (FinOps). In this context, reducing consumption directly impacts the company's business.

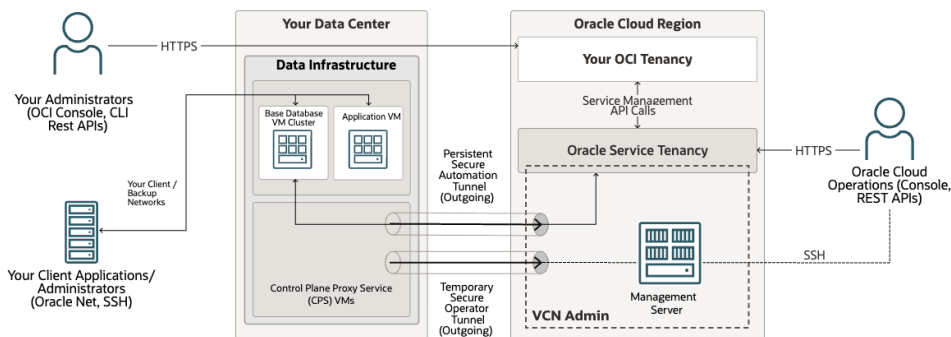
Please find all references for this chapter in the Annex.

3.5 Base Database Cloud@Customer Architecture

Guide:

This chapter and subchapters is generic and should be kept as is, please change the <CUSTOMER NAME>

3.5.1 Base Database Cloud@Customer Overview



Oracle Base Database Cloud@Customer can run Oracle Database and generic application workloads on Oracle-owned and managed infrastructure located in your data center.

Your databases run on one or more virtual machine (VM) clusters that reside on Base Database infrastructure in your data center. There are three type of VM cluster supported:

- Two node VM cluster for Database workloads
- Single node VM for Database workloads
- Single node Application VM providing an Oracle Linux operating system

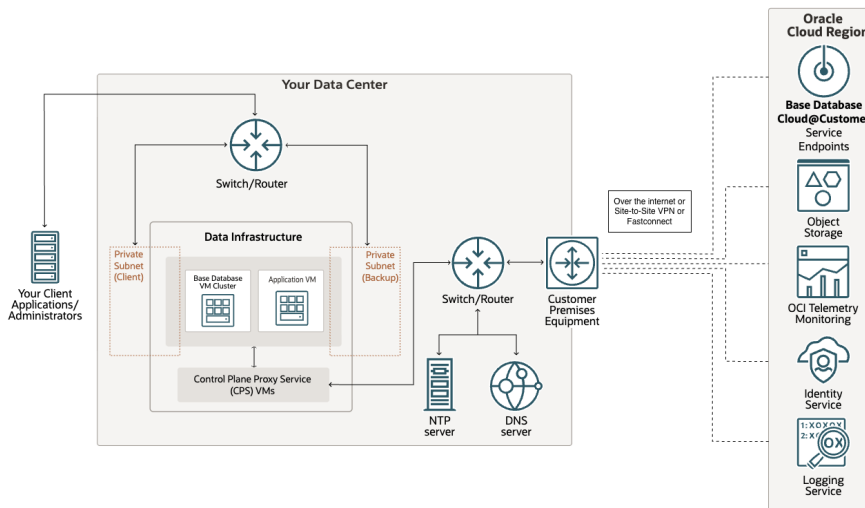
Base Database Cloud@Customer has Control Plane Server (CPS) VMs that connects to an OCI region for cloud automation and administration.

Your client applications and administrators connect to the VM clusters through client and backup networks that you create. You access your databases through standard Oracle database connection methods, such as Oracle Net. You access the VM clusters through standard Oracle Linux methods, such as token-based Secure Shell (SSH).

Your administrators can use the web-based OCI Console, command-line interface (CLI), and REST APIs to connect to your OCI tenancy over an HTTPS connection. Your tenancy sends service management API calls to the Oracle service tenancy. The Oracle service tenancy connects to the Oracle-managed admin virtual cloud network (VCN). An outgoing, persistent, secure automation tunnel connects the CPS infrastructure in your data center to the Oracle-managed admin VCN in the OCI region for delivering cloud automation commands to the VM clusters.

3.5.2 Base Database Cloud@Customer Network

Workload Requirements and Architecture



Your databases and applications run on virtual machine (VM) clusters that reside on Base Database infrastructure in your data center.

The CPS VMs running on the infrastructure are connecting to the OCI region to access Base Database Cloud@Customer service endpoints, Object Storage for staging maintenance updates, OCI telemetry monitoring, identity service, and logging service.

You can optionally connect your data center to the OCI region with FastConnect or using a VPN connection. This option uses a public virtual circuit to connect the edge of your network to the Oracle edge in the OCI region. Your edge connects to customer-premises equipment (CPE) in your data center.

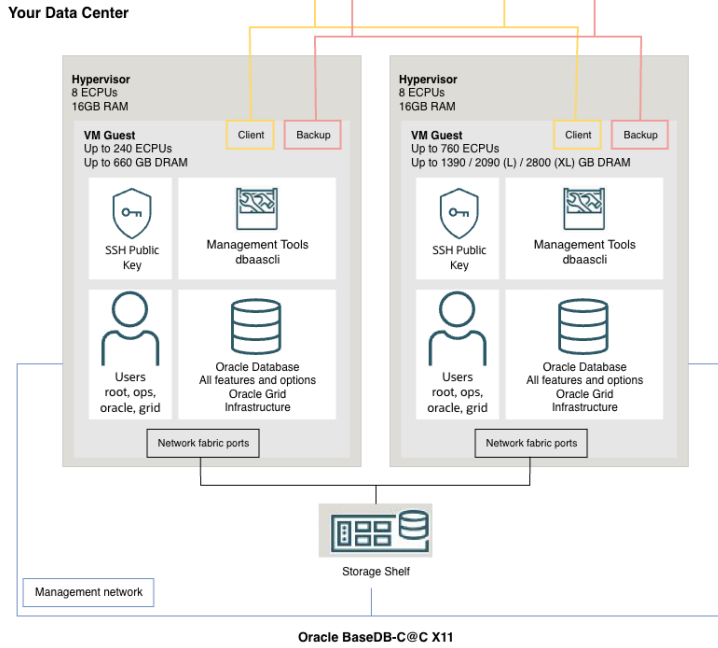
The CPE connects to a switch or router, which directs traffic to and from the CPS, network time protocol (NTP) server, domain name system (DNS) server. If you're not using FastConnect, your network should also include a proxy server (not shown in the diagram).

Your client applications and administrators connect to your VM clusters and databases over the private client and backup subnets that you access through a switch or router in your data center. You create and manage the client and backup subnets.

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3.5.3 Base Database Cloud@Customer Overview

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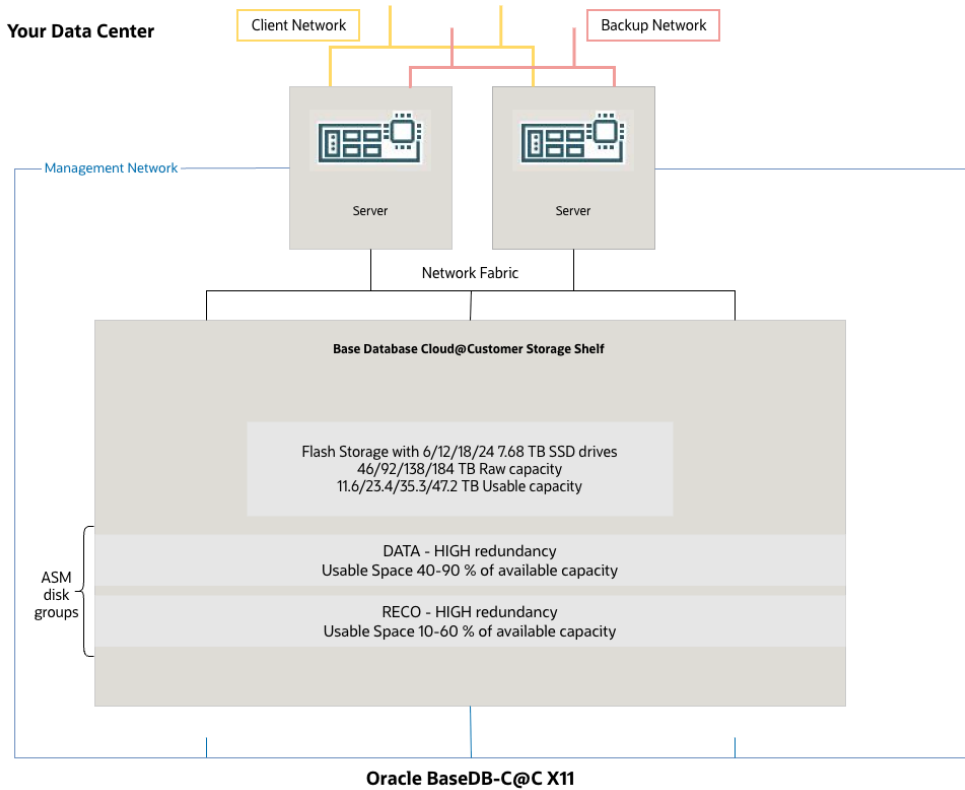
Oracle Base Database Cloud@Customer includes two servers and a storage shelf. The storage shelf is equipped with 6 7.68 TB SSD drives, which can be upgraded to 12, 18 or 24 drives.

The servers have 2 AMD EPYC 9J15 32 core 2.9 GHz (up to 4.4 GHz) processors and 768 GB. 60 processors cores and 660 GB memory are available for allocation to customer workloads per server node.

You can host up to 8 VMs per server and a maximum number of 12 two node VM clusters on the infrastructure. Within these constraints, any number of Database and Application VMs can be created. The Application VM must have the memory allocation between 1 and 8 GB for each OCPU allocated to the VM. Single node Database VMs must have a minimum of 11 GB, two node VMs a minimum of 16 GB memory allocated. The memory allocation increment for Database VMs is 1 GB.

The client and backup networks provide access to your virtual machine (VM) clusters on the servers. Oracle manages the infrastructure through the dedicated Control Plane Proxy Services VM network.

3.5.4 Base Database Cloud@Customer Storage Shelf



When you configure Oracle Base Database Cloud@Customer infrastructure, Oracle Automatic Storage Management (ASM) provisions the storage space inside the Base Database storage shelf. By default, ASM creates the following disk groups:

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The DATA disk group stores Oracle Database data files.

The RECO disk group stores the Fast Recovery Area (FRA), which is an area of storage where Oracle Database can create and manage various files that are related to backup and recovery.

Oracle recommends an 80:20 allocation, where 80 percent of storage is assigned to DATA and 20 percent is assigned to RECO.

The DATA allocation can be set from 40 percent to 90 percent. As a result, the RECO allocation ranges from 60 percent to 10 percent. A minimum of 256 GB is required for Data storage, and a minimum of 64 GB is required for Recovery Area storage.

Workload Requirements and Architecture

VM file system storage is allocated from the DATA disk group. The amount of storage required depends on the software that you plan to install on each VM. A minimum of 256 GB is required for each Database VM. VM storage is also provisioned from an Oracle ACFS volume in the DATA disk group and is required for Database Homes. A minimum of 100 GB /u01 space is required per VM.

The usable storage capacity is the storage available for Oracle Database files after considering high-redundancy ASM mirroring (triple mirroring), which provides highly resilient database storage on all Base Database Cloud@Customer configurations.

The usable storage capacity does not factor in the use of Base Database compression capabilities, which can increase the effective storage capacity.

3.5.5 Base Database Cloud@Customer Network Interfaces

Your databases run on one or more virtual machine (VM) clusters that reside on Base Database infrastructure in your data center. Base Database Cloud@Customer also includes two VMs, one on each server, to host the Control Plane Server (CPS) VMs that connect to an Oracle Cloud Infrastructure (OCI) region for cloud automation and administration.

The CPS VMs are using the outgoing, persistent, secure automation tunnel connecting to the Oracle-managed admin VCN in the OCI region through an internet access switch that you manage.

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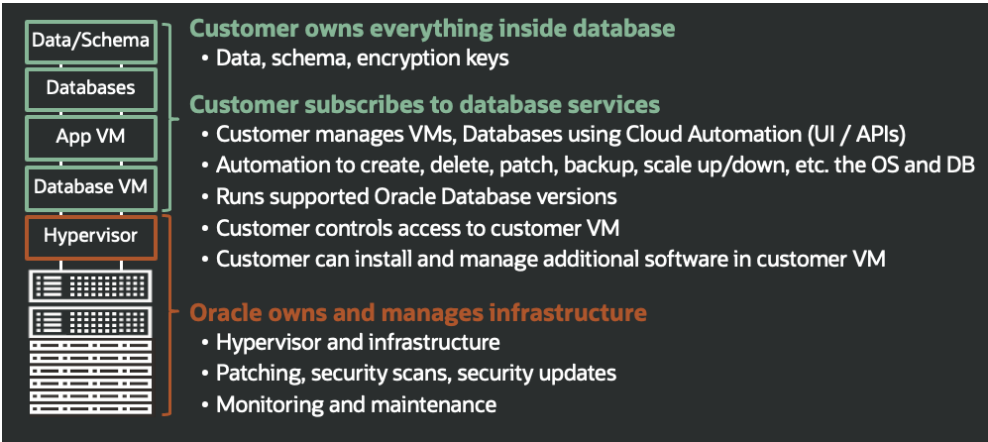
3.5.6 Explanation of Operations Service and RACI model

Guide:

This text is generic and should be kept as is, please change the <CUSTOMER NAME>

This Explanation of Operations Service describes operational processes for BaseDB-C@C including Oracle operations service scope, operations monitoring setup and telemetry data collection, issue identification and remediation, and planned/unplanned maintenance.

The picture below depicts the delineation of the roles and responsibilities between Oracle cloud operations and the end customer using the BaseDB-C@C subscription service. All BaseDB-C@C access is separated and controlled. The customer is solely responsible for maintaining appropriate security, protection, and backup of their content (e.g. applications, databases, data) within their virtual machine (VM). Oracle does not have credentials and does not access the content (applications or data) within the Customer Access Layer. Oracle has complete, exclusive access and administrative responsibilities for the Oracle Infrastructure Layer. The customer is not granted any access or administration privileges within the Oracle Infrastructure Layer.



3.5.7 Maintenance and Upgrades

Guide:

This text is generic and should be kept as is, please change the <CUSTOMER NAME>

Example:

Oracle performs updates to all Oracle-managed infrastructure components on Base Database Cloud@Customer. Oracle notifies specified contacts of infrastructure maintenance by email during infrastructure implementation. These maintenance are defined in a window to determine when quarterly infrastructure maintenance will begin, and view scheduled maintenance runs and maintenance history for your Base Database Cloud@Customer environment in the Oracle Cloud Infrastructure Console. Four maintenance sessions are run each year. The impact during maintenance or possible hardware failures varies depending on how the databases are deployed. The table below provides a summary of unavailability during maintenance.

Component	Owner	Frequency	Comments
Oracle Data Infrastructure	Oracle	Quarterly	Mandatory, servers and SSDs are updated in a rolling fashion.
Grid Infrastructure	Customer	Quarterly (recommended)	The customer decides when to apply the updates. The updates are applied in a rolling fashion.
Oracle Database Homes	Customer	Quarterly (recommended)	
Software running in the Application VMs	Customer	Follow the Application vendor recommendation	

- For databases in RAC mode with two active instances, sessions will be redirected to the second active instance.
- For databases in RAC mode with a single active instance/PDB (Singleton Service), sessions will be redirected to the second instance once the service is started on the second instance/CDB



Workload Requirements and Architecture

In these first two cases, transactions in progress on the instance that is being interrupted can be replayed if Application Continuity and Transparent Application Fail-Over are active. It should be noted that maintenance starts by stopping application services in drain mode. Without these components, there will be an outage until reconnection on the second instance.

If the database is protected by Data Guard, a switchover can be done in order to perform the upgrade with minimal impact, similar to RAC.

3.6 Sizing and Bill of Materials

Guide:

Estimate and size the physically needed resources of the Workload. The information can be collected and is based upon previously gathered capacities, business user numbers, integration points, or translated existing on-premises resources. The sizing is possibly done with or even without a Physical Architecture. It is okay to make assumptions and to clearly state them! Clarify with sales your assumptions and your sizing. Get your sales to finalize the BoM with discounts or other sales calculations. Review the final BoM and ensure the sales are using the correct product SKUs / Part Number. Even if the BoM and sizing were done with the help of Excel between the different teams, ensure that this chapter includes or links to the final BoM as well.

3.6.1 Choosing the appropriate sizing method

Guide:

Choose the appropriate sizing approach and method based on your use case. Look at the guides and examples for the appropriate method.

For Application VM sizing we must rely on input from the customer, since we have no procedures in place to conduct a sizing exercise for specific application workloads.

3.6.2 Collected Data for Sizing

Guide:

Available sizing methods:

- *Oracle Cloud Capacity Analysis (OCCA) – section 3.6.3*
- *Manual Sizing – section 3.6.4*

3.6.3 Oracle Cloud Capacity Analysis (OCCA) data collection

Guide:

With OCCA Oracle Cloud Capacity Analytics the data collection comes from Oracle Enterprise Manager, AWR (Automatic Workload Repository) and/or is manually added.

Collecting data with Oracle Enterprise Manager is preferable, since the data collection procedure is easier to execute, while the collected data can span up to one year, which helps to identify seasonal trends in the workloads. Revert to AWR Miner based collection if Enterprise Manager is not used by the customer.

Customer data is gathered with a script provided by Oracle with clear instructions on how to execute. It's customer responsibility to run the extract script and optionally obfuscate the data then upload the output to Oracle using secure methods (Oracle provides a Secure Site as part of the BaseDB-C@C sales process).

The output needs a first validation, where any anomalies in the data collection, such as errors, missing data, unrealistic outliers/deviations or unexpected growth can be identified.

This data is then grouped in Cohorts which are in the context of sizing grouping of databases together for the purpose of being deployed in VM Clusters. In the sizing process customer define Cohorts/VM Clusters for each database taking the following into consideration:

- *Network segmentation due to security*
- *Workload isolation*
- *Application / User separation*
- *Required Service levels (for example MAA Gold, Silver, Bronze)*
- *Size of the workload*
- *Physical Location*

After receiving the data collection files from the customer, follow the OCCA user guide to complete the sizing exercise. Set up the property file with the cohort names for each database based on the discussion with the customer.

Example:

The sizing exercise is based on the following inputs:

Input type	Number of Database	Time span covered
AWR Miner	8	06/01/2025 - 06/30/2025
EMCC Extract	34	06/30/2024 - 06/30/2025

3.6.4 Manual sizing

Guide:

Wherever possible we try to avoid manual sizing efforts. Since the data input is limited, that can affects the correctness of the output and. This method can only be used as a not-to-exceed type sizing, with no workload seasonality, which usually means that from all the sizing methods manual sizing can produce the least accurate results. Whenever it is possible please use a different sizing method.

Manual data collection can mean several things, for example:

- Finding the environments in scope, for example PROD, TEST, DEV, QA, etc
- Understanding the number of databases
- Versions used
- RAC or Single Instance
- Dataguard or not
- Looking at number of licenses in use by the customer
- Collecting data on existing hardware platform by interviewing customer
- Asking for customer estimates and/or metrics on CPU, memory and storage utilization in current environment
- Using backup data to estimate size of databases (querying Full Backup size from RMAN catalog)

Statspack collection based sizing should be only used in an effort to determine the workload of Standard Edition Database environments, which we want to move to an Base Database Cloud@Customer system.

Example:

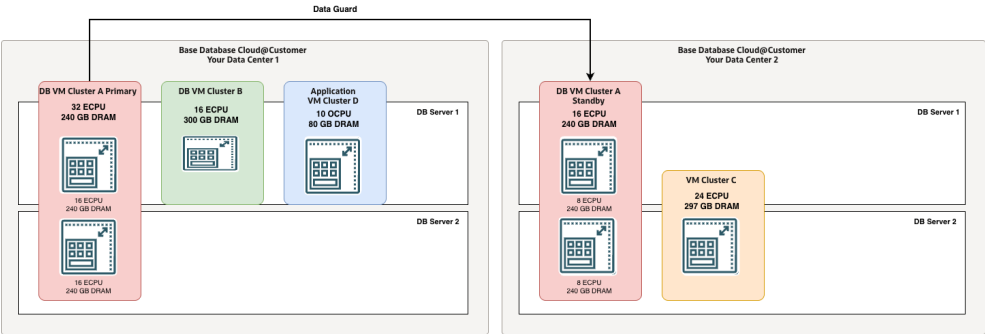
The sizing exercise is based on the following inputs:

Infrastructure	Number of Databases	CPU type	Allocated number of cores	Allocated Memory	Allocated Storage
Oracle Base Database X6-2	8	Intel Xeon E5-2699-v4	16	650 GB	24 TB
Generic x86	28	Inter Xeon Gold 6328HL	42	1390 GB	98 TB
IBM Power	10	POWER9 SMT8	20	380 GB	40 TB

3.6.5 Defined VM Clusters for Sizing

- Guide:
- Cohorts in the context of sizing are grouping databases together for the purpose of being deployed in VM Clusters. In the sizing process customer define Cohorts/VM Clusters for each database taking the following into consideration:
- Network segmentation due to security
 - Workload isolation
 - Application / User separation
 - Required Service levels (for example MAA Gold, Silver, Bronze)
 - Size of the workload
 - Physical Location

Document the VM Clusters that has been agreed with the customer and any assumptions made. If you have a more complex VM Cluster layout with VM Cluster Node Subsetting ([Overview of VM Cluster Node Subsetting](#)) then consider visualizing it. There is an example below.



Example:

Based on the customers' two different data centers and requirements for Production and non-Production environments which include different service levels as well as separate Client network segments for the environments, the following Cohorts (VM Clusters) have been defined.

Each VM cluster runs a single version of the Oracle Grid infrastructure (GI) to be chosen by the customer among the most recent. Each cluster can run different versions of Oracle Homes. Each independent VM clusters has ECPU assigned during the deploy phase of the VM cluster, so, we can assign different number of ECPU to different VM clusters and environments.

VM Clusters to be deployed are listed in the following tables together with the criteria that define why separate VM Clusters are deployed.

Your Data Center 1

Data Center 1 <NICKNAME > VM Clusters	Environment	Client Network Segment	Backup Network Segment	MAA Service Level
Production A Primary	Production	Client Network A	Backup Network	Gold
Non-Production B	Test	Client Network B	Backup Network	Silver
Miscellaneous D	Test	Client Network D	Backup Network	Silver

Your Data Center 2

Data Center 2 <NICKNAME >	Environment	Client Network	Backup Network	MAA Service
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Workload Requirements and Architecture

VM Clusters		Segment	Segment	Level
Production A Standby	Production	Client Network A	Backup Network	Gold
Non-Production C	Test	Client Network C	Backup Network	Silver

Please see the Base Database Cloud@Customer X11 Data Sheet for the latest information.

3.6.6 Sizing Assumptions

Guide:

Document the customer requirements that is included in the sizing as well as assumptions made for the sizing. This information is provided as part of the sizing exercise.

Scaling

If the data collection captured performance requirements over time (OCCA-based sizing), the impact of scaling can be calculated in the results. In that case, indicate the time utilization percentages used (e.g., 90%, 85%, 80%) in the results.

Growth

The predicted growth of the targeted workloads needs to be considered to ensure that the customer has sufficient infrastructure available and is aware of the increase of ECPU consumption over time.

This growth will be composed of an assumption of the expected organic growth of the existing target workloads, but additional new or not currently in scope workloads should also be considered.

The ramp up plan of ECPU consumption should consider the on-boarding plan of the workloads for the customer. e.g. They may start with a smaller test environment prior to starting migrations and moving into production.

DR environment sizing

Under normal operating conditions, pure DR and readable Standby databases will have lower processing demands than during a disaster. During normal operations, required vCPU and I/O consumption will typically be lower.

Single Instance to RAC conversions

When moving from Single Instance DB to a 2-node RAC DB, it's advised allocating the same amount of memory that would be allocated to the single VM to both nodes in the VM-Cluster. Hence doubling the amount of memory. When in this case a node would go down, the remaining node can run the DB activities with no performance impact.

SGA sizing – Large Pages

SGAs are often oversized. The AWR report for specific databases i.e. the SGA and PGA advisor can determine whether it is safe to reduce. For dedicated platform cloud deployments (BaseDB-C@C, BaseDB-D, BaseDB-D@Hyperscaler) it is also important to consider settings for Large Pages.

When a VM is initially created, each VM's operating system is configured with 50% of the memory allocated to the VM for large pages, and databases are configured to use that memory for their SGA. Oracle recommends that you not modify the large pages configuration unless you understand the implication of any changes you make. If you configure large pages to be greater than 60% of total memory, then the cloud automation will resize it to 60% of total memory. This automatic resize is to ensure sufficient conventional memory for the virtual machine to start.

Storage sizing

Applying or introducing compression can have a significant impact on the required storage. The compression ratio depends highly on the data-set and the selected compression algorithm. Applying compression often needs some thought work from the customer and sometimes will only be applied after DB migration. Make sure there is sufficient buffer capacity in the target platform.

It's advised to use Compression Advisor to estimate any compression ratio's applied during a sizing exercise. Document and share any assumption taken.

The following example has some useful assumptions that should be considered.

Example:

The following customer requirements and general assumptions have been used in the sizing.

- All analyses are based on previous resource consumption and does not necessarily predict the future workload behavior.
- The defined cohorts (VM Clusters) and databases can differ from the current deployment.
- The sizing computations are based on the hourly average resource consumption of databases in each cohort.
- Cohort metrics are aligned by time and then the sum of peaks is computed for each hour.

Workload Requirements and Architecture

- With OCCA based sizing the utilization percentage is indicated (90/85/80%)
- Hourly data is adjusted based on peaks found in the daily data over the past year (valid for EM data case).
- Using in-memory or other database options may reduce/increase the required processing power.
- Using different execution plans will influence the required processing power.
- For each VM Cluster, an additional 16 ECPUs were added for OS+GI. Metrics for DR DBs were considered as 20% of the corresponding Primary DBs

Infrastructure growth

The solution needs to be sized to accommodate 4 or 5 years of growth. For other Oracle Database Services, the chosen platform needs to be capable of expanding to accommodate future growth for a reasonable amount of time before re-platforming becomes necessary.

Sizing Attribute	Overall Growth Percentage
CPU Growth %	xx %
Memory Growth %	xx %
Storage Growth %	xx %
Storage IOPS Growth %	xx %
Sizing Outlook Years	4/5

Next to the infrastructure growth percentages is also wise to make sure there is a safety margin and not fill-up infrastructure for the full 100% of what is physically available/installed.

Sizing Attribute	Margin Percentage
CPU margin %	xx %
Memory margin %	xx %
Storage margin %	xx %

DR environment sizing

Standby Sizing Attribute	Normal Operation (Standby Mode)	Activated During Disaster
Database Storage	100 % of Primary	100 % of Primary
Instance Memory	100 % of Primary	100 % of Primary
Sessions	100 % of Primary	100 % of Primary
ECPU consumption	N % of Primary (default 20 %)	100 % of Primary
I/O consumption	N % of Primary (default 20 %)	100 % of Primary

Single Instance to RAC conversion

Sizing Attribute	Margin Percentage
CPU Overhead % (Overhead due to RAC cache fusion)	xx %
RAC Memory Overhead % (Overhead to handle node failures)	xx %

SGA sizing – Large Pages

Workload Requirements and Architecture

Sizing Attribute	Margin Percentage
Large Pages Memory Alloc.(50% default, 60% max)	xx %

Storage sizing

Sizing Attribute	Margin Percentage
Percentage of overall storage capacity or data files (DATA)	xx %
Percentage of overall storage capacity for redo, fra, ... (RECO)	xx %

3.6.7 Proposed Hardware Shapes

Guide:
Document the proposed hardware shape and any additions, corrections or customer requested changes made.

Example:
With the summarized sizing information applied to High Availability and Performance Best Practices both in general and specific to the Base Database Cloud@Customer, the following hardware shapes are recommended.

Datacenter	Infrastructure	Number of Storage Expansions
Primary	BaseDB-C@C X11	1
Secondary	BaseDB-C@C X11	1

3.6.8 Estimated Consumption Plan

Guide:
Base Database Cloud@Customer infrastructure runs over a 48-month commitment with the possibility for the customer to extend an additional 12 months for a total of 60 months or 5 years.
The consumption can either be covered by Bring Your Own License (BYOL), or by Licenses Included. With BYOL customer will consume BYOL ECPU and will also be required to have sufficient licenses for DB and Options (except certain options (for example TDE), which is included).
The available Database Edition Licenses for BaseDB-C@C are the following:

- Oracle Base Database Service Cloud@Customer - Standard Edition
- Oracle Base Database Service Cloud@Customer - Enterprise Edition
- Oracle Base Database Service Cloud@Customer - Enterprise Edition High Performance
- Oracle Base Database Service Cloud@Customer - Enterprise Edition Extreme Performance
- Oracle Data Infrastructure Cloud@Customer - Application Compute
- Oracle Base Database Service Cloud@Customer - All Editions BYOL

The choice of licensing model can be changed on the VM Cluster level.
The Application VM is using OCPU as the consumption metric, all the Database licenses are using ECPU.

Executing one of the supported sizing exercises we have to document the estimated consumption, which has to correlate with the Bill of Materials.

Example:

Workload Requirements and Architecture

This is the proposed Consumption Plan over five years based on the sizing, growth and other assumptions including OS/GI load.

Datacenter	Cohort	Year 1 OCPU/ECPU per hour	Year 2 OCPU/ECPU per hour	Year 3 OCPU/ECPU per hour	Year 4 OCPU/ECPU per hour	Year 5 OCPU/ECPU per hour
Primary	DB PROD	20	24	28	32	38
Secondary	App	4	6	6	6	8
Secondary	DB	12	12	16	16	20

3.6.9 Commercial & Technical Exceptions

Guide:
This is internal section for reference. Delete before creating customer version.

Data Centre	Exception Details
Data Center 1 <NICKNAME>	
Data Center 2 <NICKNAME>	

3.6.10 Bill of Materials

Guide:
You can add the Bill of Materials in any shape or format. Create a table, bullet list or a picture from the CPQ quote.
Base Database Cloud@Customer infrastructure needs to have a CPQ quote for each location it is going to be shipped to (due to Ship-to addresses and Site Details in the SAR Service Activation Request potentially being different). So, for a solution with two data centers there will be a minimum of two CPQ quotes, hence you should list that as two independent BOM.
The Base Database Cloud@Customer consumption can be either based on BYOL or using one of the Database License Included options explained in section 3.6.6. Licensing type is a parameter of the VM Cluster, for example, VM Cluster 1 can be License Included with Enterprise Edition High Performance, iwhile VM Cluster 2 can be BYOL. Correct SKU's need to be used and can be added to one of the BOM if there are several.
Example:

This is the Bill of Material (BOM) for the two data centers.

Your Data Center 1 - Bill of Material line items	Metric	Quantity
Oracle Data Infrastructure Cloud@Customer - X11	Hosted Environment Per Month	48
Oracle Data Infrastructure Cloud@Customer - X11 Storage	Hosted Environment Per Month	96
Data and Device Retention Service for Oracle Data Infrastructure Cloud@Customer	Hosted Environment Per Month	48
Data and Device Retention Service for Oracle Data Infrastructure Cloud@Customer - Storage	Hosted Environment Per Month	48
Oracle Database Infrastructure Cloud@Customer - Installation and Activation	Each	1
Oracle Base Database Cloud@Customer Database ECPU – Enterprise Edition Extreme Performance	ECPU Per Hour	1,380,400

Oracle Data Infrastructure Cloud@Customer - Application Compute	OCPU Per Hour	780,000
Your Data Center 2 - Bill of Material line items	Metric	Quantity
Oracle Data Infrastructure Cloud@Customer - X11	Hosted Environment Per Month	48
Oracle Data Infrastructure Cloud@Customer - X11 Storage	Hosted Environment Per Month	96
Data and Device Retention Service for Oracle Data Infrastructure Cloud@Customer	Hosted Environment Per Month	48
Data and Device Retention Service for Oracle Data Infrastructure Cloud@Customer - Storage	Hosted Environment Per Month	48
Oracle Database Infrastructure Cloud@Customer - Installation and Activation	Each	1
Oracle Base Database Cloud@Customer Database ECPU – Enterprise Edition Extreme Performance	ECPU Per Hour	720,000
Oracle Data Infrastructure Cloud@Customer - Application Compute	OCPU Per Hour	180,000

Annex

4.1 Security Guidelines

4.1.1 Oracle Security, Identity, and Compliance

Oracle Cloud Infrastructure (OCI) is designed to protect customer workloads with a security-first approach across compute, network, and storage – down to the hardware. It's complemented by essential security services to provide the required levels of security for your most business-critical workloads.

- [Security Strategy](#) – To create a successful security strategy and architecture for your deployments on OCI, it's helpful to understand Oracle's security principles and the OCI security services landscape.
- The [security pillar capabilities](#) reflect fundamental security principles for architecture, deployment, and maintenance. The best practices in the security pillar, help your organization to define a secure cloud architecture, identify and implement the right security controls, and monitor and prevent issues such as configuration drift.

4.1.2 References

- The Best Practices Framework for OCI provides architectural guidance about how to build OCI services in a secure fashion, based on recommendations in the [Best practices framework for Oracle Cloud Infrastructure](#).
- Learn more about [Oracle Cloud Security Practices](#).
- For detailed information about security responsibilities in Oracle Cloud Infrastructure, see the [Oracle Cloud Infrastructure Security Guide](#).

4.1.3 Compliance and Regulations

Cloud computing is fundamentally different from traditional on-premises computing. In the traditional model, organizations are typically in full control of their technology infrastructure located on-premises (e.g., physical control of the hardware, and full control over the technology stack in production). In the cloud, organizations leverage resources and practices that are under the control of the cloud service provider, while still retaining some control and responsibility over other components of their IT solution. As a result, managing security and privacy in the cloud is often a shared responsibility between the cloud customer and the cloud service provider. The distribution of responsibilities between the cloud service provider and the customer also varies based on the nature of the cloud service (IaaS, PaaS, SaaS).

4.1.4 Additional Resources

- [Oracle Cloud Compliance](#) – Oracle is committed to helping customers operate globally in a fast-changing business environment and address the challenges of an ever more complex regulatory environment. This site is a primary reference for customers on the Shared Management Model with Attestations and Advisories.
- [Oracle Security Practices](#) – Oracle's security practices are multidimensional, encompassing how the company develops and manages enterprise systems, and cloud and on-premises products and services.
- [Oracle Cloud Security Practices](#) documents.
- [Contract Documents](#) for Oracle Cloud Services.
- [OCI Shared Security Model](#)
- [OCI Cloud Adoption Framework Security Strategy](#)
- [OCI Security Guide](#)
- [OCI Cloud Adoption Framework Security chapter](#)

4.2 Maintenance and Patching

The following guidelines are recommended to reduce the operational risk and business impact during customer maintenance activities:

- Schedule updates during low-usage periods and outside business hours to minimize disruption. Ensure that patching activities across different components are coordinated to avoid scheduling conflicts.
- Whenever it is possible, execute the patching activities in this order: Base Database Infra → VM Cluster → Oracle Grid Infrastructure → Oracle Database Software → Container Database and add buffers between maintenances in case of unexpected issues.
- Always schedule non-production before production environments and follow a layered approach per environment. This enables your application teams to run functional and performance tests to identify any potential regressions before they may impact the business and to correlate any possible issue with the corresponding patching event.
- Ensure that all components scheduled for patching are operational and available before starting the update.
- Before you apply any patch, run the precheck operation to ensure your VM cluster, Grid Infrastructure, or Database Home meets the requirements for that patch.
- Maintaining system security and performance requires **regular patching**, including monthly OS updates and quarterly database and Grid Infrastructure patches.
- Use OCI-supported **Software Images**, including custom database and Grid Infrastructure images with RU and interim patches, for provisioning and patching.
- Installing OS packages outside of Oracle's official Base Database updates can create dependency issues that disrupt future updates, so Oracle advises minimizing such customizations. If you choose to install extra packages, it's best to use scripts that automate their removal and reinstallation to avoid complications during updates.
- You must use **cloud-native interfaces** like OCI Console, API, CLI, **dbaascli**, or fleet update services, instead of manual tools like opatch, unless explicitly indicated.
- Ensure you have a valid backup of your databases before you apply any patches.
- Oracle recommends out-of-place patching for databases and Grid Infrastructure, that implies moving them to a Home that uses the target patching level when possible. This technique requires less downtime and allows you to easily roll back to the previous version by moving it back to the old Home.
- Additional Oracle tools, such as Oracle Real Application Testing (RAT), can be utilized to support performance tests and replicate production database load in production-like environments.
- In mission-critical databases protected by Oracle Data Guard, standby databases are always patched first. You could convert the already patched physical standby database into a snapshot standby database and test the patch before the primary is applied. This step would reduce database protection against disaster temporarily while the tests are run. The risk can be mitigated with the use of a second standby database. The recommendation is to convert one standby to a snapshot standby and leave the other standby for protection purposes only.
- When managing updates across a large number of guest VMs, Grid Infrastructure or database deployments, it is recommended to leverage fleet maintenance capabilities, such as Base Database Fleet Update or Fleet Application Management, to streamline patching, ensure consistency, and reduce operational overhead.



Annex

- Patch activities can be monitored through the OCI Console by reviewing Work Request logs or the logs of the fleet services used for the updates. For guest VMs, Grid Infrastructure, and databases, the default component-specific logs are also available to track progress and identify issues.